BHASKARACHARYA NATIONAL INSTITUTE FOR SPACE APPLICATIONS AND GEO-INFORMATICS

WEEKLY PROGRESS REPORT (25/05/2025 – 30/06/2025)

WEEK 2

AUTOMATIC GEO-REFERENCING	**AUTOMATED GCP DETECTION AND GEO-TIFF GENERATION**

PROJECT DESCRIPTION : AUTOMATED GCP DETECTION AND GEO-TIFF GENERATION

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25/05/2025	**Re-architected into modular Google Colab execution using google.colab.files; set up two-image upload flow (img1 / img2)** **Preprocessing Notebook (Iteration 1) — Built .ipynb with 4 preprocessing methods- CLAHE, Gaussian + Bilateral Smoothing, Unsharp Mask, Contrast Stretching + Gamma Correction.**
26/05/2025	**ASCII progress bars, and show_pair() grid display; delivered notebook but encountered SyntaxError due to malformed JSON**
27/05/2025	**The notebook was broken because it was saved as raw text instead of proper JSON — rebuilding it as a Python dictionary and converting it correctly fixed the issue.** **CLAHE + Bilateral Combo Added — Identified CLAHE as best single method for keypoint detection; appended 3 new cells (divider, explanation, code) implementing CLAHE on LAB L-channel → Bilateral Filter; final notebook structure: 13 cells**
28/05/2025	**Grid-Based SIFT Implementation — Implemented 4×4 sub-grid minimum-count SIFT with dual-detector fallback; resolved visualization bottleneck caused by hardcoded 20-point drawing cap; enabled full RANSAC-validated keypoint distribution across terrain**
29/05/2025	**Perspective Warping & Mosaicking — Implemented Cell 3: homography matrix (RANSAC, 5px threshold) → cv2.warpPerspective → binary mask → bitwise canvas merge; produced dual output**
30/05/2025	**Canvas Boundary Debugging — Diagnosed off-canvas warp issue (potential negative coordinate shift or inverted upload order); planned next-step fix: automatic bounding-box evaluation to dynamically expand output canvas regardless of image upload order** The stitched image wasn't showing up properly — the merged patch was likely landing outside the visible frame, so the next step is to auto-resize the canvas to always fit everything in view.

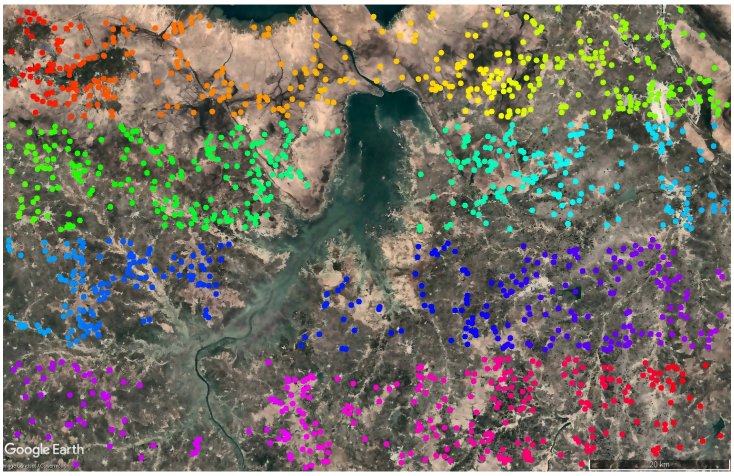
Next Week Plan	**Fix canvas boundary alignment with automatic coordinate bounding-box expansion; test on multiple image pairs; begin Step 2 keypoint matching pipeline.** **complete keypoints detection adn produce geotiff based off that.**

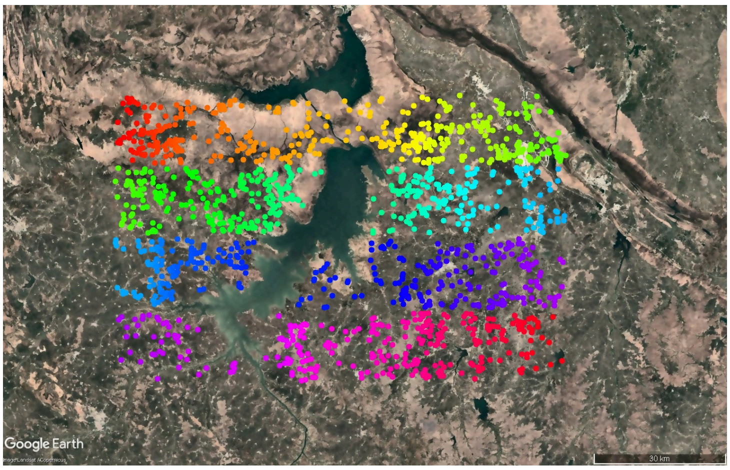
screenshots

Combo #1: CLAHE + Bilateral Filter

actual image:

image1 and 2. lake





REFERENCE:

https://docs.opencv.org/4.x/da/df5/tutorial_py_sift_intro.html

https://docs.opencv.org/4.x/dc/dc3/tutorial_py_matcher.html

https://docs.opencv.org/4.x/d9/dab/tutorial_homography.html

https://colab.research.google.com/